

US/HK Signal Robot Rulebook (Discussion Draft)

Date: 2026-06-09

Scope: Rule-based technical signal engine for discretionary trading support (no auto-order execution)

1) Objective and Operating Mode

- Purpose: generate structured trade-reference signals, then push reminders to Feishu/Telegram.
- Role boundary: decision support only; final execution is manual by trader.
- Runtime mode now:
- Background monitor loop: ``python run.py``
- On-demand Q&A helper: ``python ask.py price SYMBOL`` / ``python ask.py advice SYMBOL``

2) Current Live Universe and Timeframes

- Watchlist (active):
- HK.07709
- US.SPY, US.QQQ, US.MU, US.SNDK, US.GLW, US.IREN, US.TSLA, US.MSFT, US.NVDA
- Multi-timeframe stack:
- Execution timeframe: 15m
- Trend filter timeframe: 1h
- Confirmation timeframes: 5m and 1m

3) Indicator Inputs (Per Symbol, Per Poll)

The engine computes:

- MA20, MA50
- RSI(14)
- Recent swing high / swing low (lookback 20 bars)
- Volume ratio = latest volume / 20-bar average volume
- ATR(14) and ATR% (ATR / close)

Minimum data safety requirement:

- At least 60 bars required for full indicator set.

4) Signal Taxonomy and Priority

Possible signal types:

- ``exit_warning``
- ``breakout_candidate``
- ``long_setup``
- ``observation_only``

Evaluation priority order (first match wins):

1. `exit_warning`
2. `breakout_candidate`
3. `long_setup`
4. `observation_only`

Quality scoring:

- Score mapped to evidence strength:
- ≥ 80 : strong
- ≥ 60 : medium
- < 60 : weak
- Score mapped to priority tier:
- A (≥ 80), B (≥ 60), C (< 60)

5) Rule Definitions (Trigger Logic)

5.1 ``long_setup`` (bullish setup)

All conditions must pass:

- Price > MA20 and Price > MA50 on 15m
- MA20 > MA50 OR MA20 rising vs previous bar
- RSI recovers upward through threshold:
- previous RSI < 48 $\leq$ current RSI
- 1h trend supportive:

- trend price > trend MA20 and trend MA20 >= trend MA50
- Price not too close to swing resistance:
- resistance distance > 0.8%
- Lower timeframe confirmation votes:
- at least 1 of (5m, 1m) has price >= MA20 and RSI >= 50
- ATR filter pass (if enabled)
- Relative strength filter pass (if enabled)

Output fields include:

- Suggested entry zone around current price (+/-0.4%)
- Invalidation = min(MA20, swing low)
- Nearest resistance = swing high

5.2 `breakout\_candidate` (breakout watch/entry candidate)

All conditions must pass:

- Price > recent swing high (breakout level)
- Volume ratio >= 1.2
- 1h trend not weak (trend price >= trend MA20)
- Not over-extended:
- extension <= 3.0% from breakout level
- At least one lower timeframe keeps structure (price >= MA20)
- ATR filter pass (if enabled)
- Relative strength filter pass (if enabled)
- Breakout structure pass (if enabled):
- fresh break (prev close <= breakout < now), OR
- controlled retest-hold with tolerance 0.4%

Output fields include:

- Entry zone from breakout level to current price
- Invalidation near MA20
- Dynamic resistance = current price * (1 + 3.0%)

5.3 `exit\_warning` (risk control warning)

Triggered when ANY condition is true:

- Price loses MA20 and RSI is still weakening
- RSI below exit threshold (42)
- Price breaks below recent swing low

Risk level is marked high by design.

5.4 `observation\_only` (no actionable trigger, monitor only)

Triggered when:

- Market is improving (price > MA20 OR RSI rising), AND
- Higher timeframe not clearly broken (trend price >= trend MA50)

Interpretation: conditions are not fully confirmed; avoid rushing entries.

6) Filters and Suppression Layers

6.1 ATR filter

- Enabled: true
- Pass band: 0.3% <= ATR% <= 5.0%

6.2 Relative Strength filter

- Current setting: disabled (`ENABLE\_RELATIVE\_STRENGTH\_FILTER=false`)
- If enabled, rule is:
- symbol return over lookback >= benchmark return + excess threshold

6.3 Duplicate/cooldown suppression

- Cooldown window: 30 minutes for same symbol + same signal_type
- Price-change suppression:
- if price change < 0.3% vs prior same-type signal, suppress repeat

6.4 Tier gating

- Minimum tier for sending: C (currently all tiers allowed)
- If raised to B/A, lower-tier signals are recorded but not pushed.

6.5 Symbol merge alerts

- Enabled: true
- Merge window: 20 minutes
- If a symbol flips signal type within window, system sends merged update instead of isolated duplicate-style messages.

7) Alert and Report Delivery Rules

Primary notifier:

- Channel: Feishu webhook
- Feishu keyword protection:
- all outgoing text is prefixed with `[signal-assistant]` if missing
- avoids keyword-check rejection errors

Message payload content:

- Symbol, timeframe, current price
- Entry zone, invalidation, resistance
- Risk level, ATR%
- Quality score + tier + evidence strength
- Rule explanation
- Optional LLM analysis block (when enabled)
- Disclaimer

Report reminders:

- Daily report: enabled, 16:10 (US market timezone), top 5 symbols
- Weekly report: enabled, weekday=4, 16:20, top 8 symbols

8) LLM Explanation Layer (Optional)

- Current state: disabled (`ENABLE\_LLM\_ANALYSIS=false`)
- When enabled:
- only selected signal types use LLM (default long_setup, breakout_candidate)
- LLM does not trigger trades; it explains existing rule-based signals.

9) Polling, Session Control, and Storage

- Poll interval: 60 seconds
- Trading-hours-only gate: false (engine runs cross-session currently)
- Database: `data/signals.db`
- `signal\_history`: all generated signals with payload JSON
- `notification\_logs`: send status/errors

10) Current Practical Interpretation (For Expert Review)

- The framework is currently conservative on new long entries and relatively sensitive to short-term weakness warnings (example: repeated SPY exit_warning under cooldown control).
- Practical knobs to discuss with your stock specialist:
- RSI thresholds (entry recovery 48 / exit 42)
- breakout volume threshold (1.2)
- extension cap (3.0%)
- cooldown and duplicate tolerance (30 min / 0.3%)
- whether to lift minimum send tier from C to B
- whether to enable relative strength filter in live usage

11) Compliance and Risk Statement

- This robot is not an execution system and does not place orders.
- All outputs are reference signals for discretionary decision support only.
- Not financial advice.